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Optimizing Traffic Flow Using Data-
Driven Simulation Models with Rain-
Adaptive Pedestrian Priority

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Contents

1	Abstract	6
2	Introduction.....	7
2.1	Background	7
2.2	Problem Statement.....	9
2.3	Aim and Objectives	10
2.4	Significance of the Study	10
2.5	Ethics	12
3	Literature Review	13
3.1	Theoretical Background and Smart Traffic Control	14
3.2	Rainfall Effects on Urban Mobility.....	15
3.3	SUMO Traffic Modelling	16
3.4	Data-Driven and Machine Learning Approaches	18
3.5	Smart Cities and Policy Alignment	18
3.6	Real-Time Systems and Roadside Infrastructure.....	19
3.7	Limitations in Existing Literature	19
3.8	How the Literature Informs this Project.....	20
3.9	Summary	21
4	Methodology.....	22
4.1	Methodological Strategies Considered.....	22
4.2	Methodology Justification	22
4.3	Methods and Techniques	23
4.4	Methodology Limitations	24
4.5	Tools Used.....	24
4.6	Summary	25
5	Design and Development.....	26
5.1	System Design and Specification.....	26
5.2	Development Process	27
5.3	Data Gathering	28
5.4	System Inputs and Data Sample Characteristics	29
5.5	Consideration of this Technique.....	29
5.6	Summary	31

6	Implementation and Testing	32
6.1	Project Implementation.....	32
6.2	Translating Conceptual Logic into Code.....	32
6.3	Testing and Validation	33
6.3.1	Pedestrian walk time and Vehicle speed	34
6.3.2	Pedestrian delay comparison.....	34
6.3.3	Vehicle flow comparison.....	34
6.4	Rain vs Rain-adaptive Scenarios Comparison.....	35
6.4.1	Total Emission Comparison.....	36
6.4.2	Pedestrian Movement Efficiency	36
6.4.3	Pedestrian Maximum Speed Distribution.....	37
6.4.4	Pedestrian Time Loss	37
6.4.5	Pedestrian Trip Durations	38
6.4.6	Average Speed vs Route Length	38
6.4.7	Speed Factor Distribution	39
6.4.8	Time Loss Distribution	39
6.4.9	Vehicle Trip Duration.....	40
6.4.10	Waiting Time Distribution.....	40
6.5	Summary	41
7	Discussion and Critical Appraisal of Results	42
7.1	Comparison Within Existing Research	42
7.2	Interpreting Key Findings	42
7.2.1	Improved Pedestrian Experience	42
7.2.2	Vehicle Efficiency.....	43
7.3	Environmental Benefits	43
7.4	Statistical Significance	43
7.5	Innovation from Past Methods.....	44
7.6	Limitations and Reflection.....	45
7.7	Summary	45
8	Conclusion and Recommendations.....	46
8.1	Key Findings	46
8.2	Project Performance	47
8.3	Reflection and Skills Learned	47

8.4	Recommendations	48
9	References.....	49
10	Appendix.....	51
10.1	Guide to Run Simulations.....	51
10.2	Ethics Training Certificate	51
10.3	Rain Scenario Generation Script.....	51
10.4	Rain-adaptive Scenario Generation Script	53

1 Abstract

This project presents the design, simulation, and analysis of a rain-adaptive pedestrian priority system aimed at enhancing urban mobility and pedestrian safety during adverse weather conditions. Using the SUMO (Simulation of Urban Mobility) platform, the system dynamically adjusts pedestrian signal timings and vehicle speeds based on real-world historical rainfall data. Python scripting, TraCI API, and Jupyter Notebook analysis enabled the development of both standard (rain) and rain-adaptive traffic scenarios, with a focus on Trafalgar Square, London, England. Comparative testing demonstrated significant improvements: pedestrian delays were reduced, vehicle throughput increased, and emissions notably decreased under rain-adaptive control compared to non-adaptive rain conditions. The integration of real rainfall data into signal behaviour, without relying on live sensor networks, positions this system as a scalable, low-cost proof of concept for intelligent traffic management. Findings align with prior research advocating environmental sensitivity in adaptive traffic systems and demonstrate that even simple, rule-based dynamic controls can substantially improve efficiency, sustainability, and safety in urban areas. Recommendations for future work include real-time data integration and expansion across larger network models.

2 Introduction

2.1 Background

Urban traffic congestion is a significant global issue, increased by rapid urban population growth, rising motor vehicle usage, and variances between traffic supply and demand(Wu, 2022). Traffic congestion, pedestrian safety, and the adverse consequences of weather, particularly rainfall, are some of the most urgent problems facing urban transport today. Urban junctions are frequently the focal points of delays, traffic, and safety hazards since they are the meeting places of pedestrian and vehicular movement(Gheorghe and Soica, 2025a).

Urban expansion presents challenges for traditional traffic management systems, which frequently lack the capacity to respond effectively to changing environmental conditions and situational factors, including adverse weather events like rainfall. Traditional traffic light systems, still widely in use, typically operate on fixed timers that do not account for real-time traffic flow or environmental variables. This inflexibility leads to inefficient signal timings, worsened traffic jams, and longer wait times for both vehicles and pedestrians(Sun and Liu, 2021), These algorithms frequently ignore real-time weather and road data, which results in inefficiencies including increased accident risk and longer pedestrian wait times. Rainfall has been found to influence driver behaviour and vehicle performance in addition to lowering road surface friction and visibility, which frequently leads to an increase in crash frequency and traffic delays(Jung, Qin and Noyce, 2011).

Researchers have looked into ways to address these issues by designing real-time intelligent traffic systems using computer vision, deep learning, and fuzzy logic-based control. (Vaghela and Solanki, 2023)suggest a model that uses video cameras to gather real-time traffic data, applies deep learning for object recognition, and uses fuzzy inference to determine the best signal timing. Their approach shows that adaptive models can outperform fixed-timing systems in terms of responsiveness and efficiency when they are based on accurate traffic classification. Rain can increase the complexity of traffic conditions by diminishing visibility, lowering vehicle speeds, and increasing pedestrian vulnerability at road crossings. These challenges require intelligent and adaptive solutions that improve urban mobility while emphasising safety. Despite international attempts to enhance traffic flow, the majority of urban traffic systems continue to be reactive and vehicle-centric, with little ability to adapt to the changing weather conditions in real time. Rain, on the other hand, drastically changes traffic patterns by slowing down vehicle speeds, increasing stop-and-go action, and making crossing more dangerous for pedestrians. According to the TomTom Index, average car speeds significantly decrease during rainy weather, and delays might rise by as much as 30%(‘tomtom traffic

index', 2024). The majority of urban networks, however, do not have adaptive measures in place to safeguard susceptible pedestrians in these situations.

The development of smart city infrastructure has led to the emergence of intelligent transport systems (ITS), which represent a significant shift in urban mobility by utilising data, automation, and adaptive controls to reduce congestion and enhance transport efficiency (Wu, 2022). The smart traffic management system is fundamental to the ITS framework, incorporating real-time monitoring, predictive analytics, and adaptive traffic control strategies to dynamically manage vehicle and pedestrian flow. These systems have shown considerable effectiveness in multiple countries and regions in alleviating congestion, optimising traffic signal timings, and enhancing coordination among transportation modes.

One innovative solution to these constraints is the incorporation of Cyber Physical Systems (CPS) into urban traffic signal regulation. By combining networking, computation, and physical processes, CPS frameworks allow for high-fidelity simulation, adaptive control, and real-time data collecting (Zhang *et al.*, 2020). This integration is embodied in a novel architecture presented by the Urban Traffic Signal Control - Cyber Physical System (UTSC-CPS), which provides adaptive management capabilities, flexible control strategy testing, and real-time simulation. In order to create a highly responsive, scalable, and secure traffic signal control system, it makes use of edge computing, private cloud computing, and intelligent gateways.

The foundation for extending intelligent traffic systems to address pedestrian safety, particularly in fluctuating weather conditions, is provided by UTSC-CPS, despite its primary focus on vehicle-centred optimisation. In order to balance traffic efficiency and pedestrian protection in rainy weather scenarios, the current study expands on this basis by integrating rain-adaptive pedestrian priority into traffic signal simulations using SUMO.

Furthermore, the demand for rain-sensitive adaptive methods has never been greater due to the increase in urban flooding and waterlogging caused by inadequate drainage systems, particularly in emerging cities (Hussain, Ahmed and Ali, 2018). Simulation-based control models that give vulnerable road users, such as walkers, priority during inclement weather can greatly minimise rainfall-induced traffic delays and queue lengths (Phanse, Chaturvedi and Srivastava, 2022). Despite advancements in global smart traffic innovations focussing on efficiency, a significant gap persists in adaptive response mechanisms specifically designed for weather-sensitive pedestrian safety. Rain-related delays and hazards are often overlooked in traditional ITS frameworks.

There is an increasing demand for rain-adaptive simulation models that emphasise pedestrian safety while maintaining vehicular throughput.

2.2 Problem Statement

Despite advancements in technology, adaptive traffic systems today frequently prioritise vehicle optimisation over pedestrian priority, especially during inclement weather. Due to decreased visibility and longer stopping distances, rain has a statistically significant effect on traffic accidents, especially multivehicle and rear-end collisions(Jung, Qin and Noyce, 2011). However, real-world solutions still mostly lack signal timing techniques that dynamically prioritise pedestrian movement in the event of rain.

Furthermore, local environmental factors like waterlogging, which impair pedestrian crosswalk usefulness and necessitate longer crossing times, are not taken into consideration by the majority of current simulations and adaptive models(Hussain, Ahmed and Ali, 2018; Phanse, Chaturvedi and Srivastava, 2022). Real-time adaptation for weather-related pedestrian safety is rarely incorporated, even in contemporary systems that use artificial intelligence or reinforcement learning(Wu *et al.*, 2025).

The majority of urban transport systems continue to be inflexible in the face of unpredictable weather, even with the development of smart mobility technologies. Excessive rain can lengthen wait times for pedestrians, slow down traffic, and increase the chance of collisions at intersections. The intelligent adaptability to change pedestrian signal timings or traffic light behaviour in response to rain intensity is absent from current models. This rigidity risks the safety of vulnerable road users in addition to making traffic worse. The TomTom Traffic Index reports persistent peak-hour congestion in cities like London, New York, and Paris, with additional delays during rainfall events(‘tomtom traffic index’, 2024). Yet, existing simulations and signal control strategies rarely account for how weather like rain directly impacts pedestrian wait times and crossing safety. According to (Vaghela and Solanki, 2023), a majority of urban traffic signals still operate on fixed schedules, resulting in unnecessary delays and increased congestion, especially during peak hours.

One of the main obstacles to advancement in this area, according to(Zhang *et al.*, 2020), is the gap between theoretical control strategies and their actual application. This is addressed by their UTSC-CPS model, which makes it possible to create and evaluate novel tactics using simulation before implementing them. However, there is still room for improvement in this model when it comes to the incorporation of environmentally adaptable pedestrian rules, including rain-aware crosswalk prioritisation.

2.3 Aim and Objectives

In order to improve pedestrian safety and traffic efficiency in urban environments, this study intends to create a data-driven traffic simulation model that includes rain-adaptive pedestrian priority. Simulation of Urban Mobility (SUMO) will be used to build the simulation, which will modify traffic signal timing in response to historical and current rainfall data.

In order to do this, the objectives that follow have been established:

- Create a realistic urban traffic simulation model that incorporates pedestrian and vehicle dynamics using SUMO.
- Install an adaptive traffic light system that modifies the amount of time that pedestrians must wait to cross depending on the rate that rain is falling.
- Use open APIs to incorporate weather data and dynamically modify simulation behaviour.
- Using key performance indicators (KPIs) including pedestrian wait times, traffic flow efficiency, and congestion levels, compare how well rain-adaptive and non-adaptive traffic situations perform.
- Providing insights into how weather-aware adaptive strategies can be incorporated into future smart traffic systems for safer and more efficient urban mobility.

2.4 Significance of the Study

Modern cities could become safer and more efficient if rain-adaptive pedestrian priority systems were incorporated into urban traffic simulation. Given that rain-related collisions are responsible for up to 37% of all weather-related traffic fatalities and injuries (Jung, Qin and Noyce, 2011), it is essential that traffic signals be modified to react instantly.

Additionally, as evidenced by extensive simulations conducted throughout Chinese cities, adaptive traffic systems driven by big-data analytics and machine learning have shown quantifiable gains in travel time and pollution reductions (Wu *et al.*, 2025). When combined with real-time data platforms like OpenStreetMap and Gaode, these technologies can save carbon emissions by 16% and journey times by up to 11% (Wu *et al.*, 2025).

This study adds to the larger conversation on intelligent traffic management systems by coordinating with international initiatives in the development of smart transportation. By employing traffic control systems that modify lane usage, signal timings, and vehicle prioritisation based on real-time data, numerous cities across the globe have shown the effectiveness of ITS (Wu, 2022). To ease traffic and improve commuter experiences, for instance, cities

like Tokyo and Berlin have included real-time road information updates and sophisticated traffic guiding systems(Wu, 2022). Furthermore, (Zhang *et al.*, 2020) describe how control strategy APIs and parallel simulation systems provide a reliable platform for testing and validating adaptive algorithms prior to field deployment. The focus on data-driven, scalable, and self-adaptive systems highlights the study's importance in developing intelligent, human-centered, and climate-conscious urban traffic management solutions.

The work by (Sun and Liu, 2021) demonstrates the potential of intelligent traffic light control systems that respond to real-time intersection conditions using embedded systems and simulation tools. Their system, built around a simple, cost-effective microcontroller platform, adapts signal timing based on traffic flow data, showcasing the feasibility of implementing adaptive signal logic in urban environments. This work expands on existing worldwide approaches by adding weather awareness as a new component to smart traffic models. Rain-adaptive pedestrian priority adds environmental reactivity to ITS, expanding its functionality and ultimately making urban mobility systems safer and more inclusive.

From the perspective of simulation, SUMO offers an adaptable and open-source setting for simulating these kinds of systems. As (Phanse, Chaturvedi and Srivastava, 2022) showed, it is feasible to develop validated and realistic scenarios that represent urban rain-related disturbances by incorporating rainfall-sensitive characteristics into SUMO. Furthermore, simulated tests conducted in Karachi demonstrated that during periods of intense rainfall, rain-related road ponding can cause delays of up to 35 minutes, highlighting the practical significance of this research(Hussain, Ahmed and Ali, 2018).

This study contributes a new and socially relevant layer to the body of work in intelligent transportation systems by emphasising pedestrian safety in addition to traffic efficiency. This study is to offer insights that can guide the development of weather-resilient, intelligent urban transportation frameworks, in accordance with Sustainable Development Goals (SDGs) pertaining to inclusive infrastructure, climate adaptation, and sustainable cities.

2.5 Ethics

Before this study started, the necessary Ethics Training for postgraduate research was properly finished in accordance with institutional research guidelines. The researcher gained a thorough awareness of ethical aspects from this training, particularly those pertaining to informed consent, data protection, and the appropriate use of secondary or simulated data. Additionally, a comprehensive Ethics Approval Form (with ethics number: ETH2425-4371) that outlined the project's scope, methodology, and ethical protections was filled out and formally submitted to the university's ethics committee. This stage makes sure that every facet of the study complies with the ethical guidelines established by the university, especially those that include the use of simulation tools and secondary data sources. Throughout the study, the researcher is steadfast in their commitment to respecting the values of integrity, transparency, and accountability.

3 Literature Review

Among the most important and interconnected issues facing contemporary cities are urban traffic congestion, pedestrian safety, and climate-resilient transportation. Traditional traffic management methods, which are mostly dependent on fixed signal timings, have proved insufficient as populations and vehicle numbers rise. Traffic patterns and pedestrian behaviour are made more difficult by these antiquated systems' inability to adjust to weather factors like rainfall in addition to managing dynamic vehicle flows.

In response, engineers and urban planners have created intelligent transportation systems that use adaptive logic, real-time communication, and data-driven insights to handle challenging mobility situations. Among these, intelligent traffic signal management systems are essential for increasing pedestrian safety, cutting emissions, and optimising intersection efficiency. In order to critically reflect on how these insights have directly influenced the design and implementation of this study's artefact; a rain-adaptive pedestrian signal priority model simulated in SUMO, this literature review examines both classic and modern research on smart traffic systems, simulation modelling, and rain-sensitive signal control.

As cities grow in size and infrastructure, urban mobility becomes a more complicated problem. Traffic signal optimisation has emerged as a key component in enhancing urban transportation systems, given the concerns around traffic, road safety, and environmental sustainability. Traditional traffic management systems can now be transformed into intelligent, adaptable frameworks using a variety of technologies, such as machine learning, artificial intelligence (AI), the Internet of Things (IoT), and predictive analytics. These new solutions are expected to decrease emissions, cut down on delays, and enhance traffic flow efficiency overall, especially in unpredictable situations like rain or high foot traffic. This study of the literature offers a thorough examination of the state of the art in the field of adaptive traffic control systems (ATCS), examining its advancement, efficiency, and applicability to the deployment of a simulation model that prioritises pedestrians and is rain-aware.

As populations and vehicle volumes increase, modern cities face several major difficulties, including pedestrian safety, urban traffic congestion, and climate-resilient mobility. The development of intelligent traffic signal control and smart transport systems has accelerated internationally in response to these problems. The design and implementation of the study's artefact, a rain-adaptive pedestrian signal priority system simulated in SUMO, are directly informed by recent research, which is reviewed in this chapter along with important theoretical, empirical, and technical contributions from the literature.

This study takes into account different viewpoints, unsolved issues, and useful lessons from industry deployments in addition to highlighting studies that support the approaches employed. It looks at how the project's direction was influenced and improved by pre-existing frameworks in machine learning, simulation, traffic control, and smart city infrastructure.

3.1 Theoretical Background and Smart Traffic Control

The shortcomings of fixed-time signal control models gave rise to the idea of intelligent traffic systems (ITS). Early studies by (Wu, 2022) highlighted how static signal layouts are inadequate for managing erratic traffic, particularly during peak hours or abrupt weather changes. In order to dynamically coordinate road users, Wu promoted a layered architecture in traffic management that combines sensing, modelling, and decision-making. His research has been crucial in defining how pedestrian behaviour and environmental data can be combined into a single system through adaptive traffic control.

The superior performance of adaptive traffic control systems (ATCS) over conventional time-of-day signal plans in both recurring and non-recurring traffic situations is demonstrated in one of the seminal works in adaptive signal control (Stevanovic, Dakic and Zlatkovic, 2017). These systems are especially effective in situations with unforeseen congestion since they instantly modify signal timings in response to traffic needs. According to the study's findings, adaptive systems produced more noticeable benefits during one-time occurrences like bad weather or incidents, which suggests that they could be in line with the objectives of this dissertation, particularly when it comes to pedestrian prioritisation caused by rain.

In a similar way, (Liu, Zeng and Jiang, 2017) created a real-time vehicle-counting system that modifies traffic patterns by utilising camera data. Their model achieved over 99% accuracy under a variety of scenarios by combining adaptive switching between virtual loops and detection lines based on congestion state with background subtraction. This method demonstrates how adaptive detection techniques can be used to improve traffic responsiveness, including managing pedestrian crossings in a range of traffic densities.

(Cheng, 2018) suggested a reliable highway traffic flow prediction system designed especially for rainy weather as part of an investigation into real-time monitoring under challenging circumstances. Instead of eliminating rain effects from video streams, which frequently skews valuable information, the system used adaptive clustering and graph-based flow mapping to immediately analyse footage impacted by raindrops. Its resilience to rain is in line with the simulated

settings that this project aims to address, where rain has a major impact on how vulnerable road users respond to traffic signals.

The foundation of the digital transformation of urban mobility is made up of smart traffic systems. Traditional traffic signals based on static timings are fundamentally insufficient in solving urban congestion, particularly when external variables like rainfall are involved, according to Wei Wu's foundational study from (Wu, 2022). Data collecting, traffic modelling, and adaptive control, all crucial components in the architecture of contemporary simulation-based systems were combined in Wu's study to frame smart traffic management as a multi-layered ecosystem.

Similarly, the idea of Cyber-Physical Systems (CPS) in urban traffic control was presented by (Zhang *et al.*, 2020). The modular control logic used in this project's design was inspired by their UTSC-CPS model, which highlighted the integration of cloud computing, real-time simulation, and intelligent gateways. Real-time responsiveness, a crucial component of rain-adaptive systems where environmental data and pedestrian activity must be handled dynamically, was theoretically justified by CPS-based signal control.

But not every strategy stresses intelligence as the sole way ahead. Traditionalists contend that low-density communities can still benefit from simpler traffic actuation that uses simple loop detectors. This opposing viewpoint, which is represented in more outdated systems such as SCOOT and SCAT (Lee and Chiu, 2020), draws attention to worries regarding the expense and intricacy of real-time adaptive systems, particularly in areas with inadequate infrastructure or erratic data sources.

3.2 Rainfall Effects on Urban Mobility

Rain's detrimental impact on traffic dynamics have been extensively researched. One of the most important studies on crash frequency and severity under rainfall was conducted by (Jung, Qin and Noyce, 2011), who found statistically significant increases in accident rates and vehicle delays. These findings supported the necessity of environmentally conscious traffic systems that prioritise road safety over flow optimisation, particularly for pedestrians who are more vulnerable in the event of rain.

The results of (Mishra *et al.*, 2018), who used modified Krauss car-following algorithms to estimate traffic flow during rain, provide additional evidence of the impact of rainy weather on road traffic. Rain decreases vehicle speed, road capacity, and driver visibility, as shown by their simulation data gathered from in-depth field observations. This underscores the need for dynamic signal control systems that put safety and mobility first during such situations. These

realisations form the foundation for incorporating rain awareness into this project's simulation design.

(Phanse, Chaturvedi and Srivastava, 2022) used modified parameters in the Krauss car-following model to replicate vehicle behaviour in rainy weather. Their research showed that adding rainfall intensity to simulation conditions could significantly increase the realism of traffic simulations based on SUMO. This method influenced the simulation logic of the project, especially the dynamic modification of pedestrian crossing times during periods of heavy precipitation.

(Hussain, Ahmed and Ali, 2018) added a developing country perspective by analysing the effects of floods caused by rain in Karachi, where waterlogging drastically changed pedestrian wait times and intersection usability. Their results supported the study of this project's choice to use rain-aware pedestrian logic, which takes into account both the accessibility of crosswalks and the intensity of rainfall.

3.3 SUMO Traffic Modelling

For simulating intricate traffic situations across extensive urban road networks, one of the most popular open-source programs is the Simulation of Urban Mobility (SUMO) platform. The German Aerospace Centre (DLR) created SUMO, a small, multimodal simulation environment that gives researchers a great degree of control over simulating individual vehicle and pedestrian behaviours, traffic signals, and ambient variables (Gudwin, 2016).

The adaptability of SUMO is one of its main benefits. Every car or pedestrian is seen as a unique agent with distinct characteristics and behaviours in microscopic traffic modelling, which it supports. This degree of detail makes it ideal for simulating real-time adaptive traffic regulation, particularly when climatic factors like rainfall are included because they have varying effects on individual travel patterns. SUMO facilitates multimodal traffic, such as automobiles, buses, trains, pedestrians, and cyclists, as described by (Gudwin, 2016). This research made use of these qualities to create an inclusive signal control system that gives pedestrians priority in rainy weather.

Additionally, the roadmap paper underscores the platform's capacity to incorporate OpenStreetMap (OSM) real-world maps, allowing simulations to be grounded in actual geographic layouts. This capability allowed for the import of a portion of Westminster, London, which was then altered to meet the requirements of a simulation environment that is sensitive to foot traffic for the purposes of this study. The intersection layout was created and improved using the NETCONVERT and NETEDIT tools, and random trip generators assisted in

simulating car and pedestrian movement under controlled conditions.

A notable aspect of SUMO that (Gudwin, 2016) highlights is its TraCI (Traffic Control Interface) protocol. TraCI makes it feasible to operate traffic lights, record sensor data, and dynamically modify simulation behaviour by enabling real-time communication between the simulation and external control systems using Python, Java, or C++. This feature directly supports the objectives of the project used in this study, which dynamically modifies pedestrian crossing durations and signal behaviour based on environmental input (rain).

Additionally, the platform allows for the addition of virtual sensors that may be configured to mimic actual data collection, such as Multi-Entry/Multi-Exit Detectors, Lane Area Detectors, and Inductive Loop Detectors. This was important for modelling traffic flow metrics, identifying congestion, and utilising average route lengths, waiting times, and time loss data to validate artefact performance.

Additionally, SUMO promotes organised simulation setup by using XML-based files for routes, network topology, extra parameters (such traffic lights), and output logging. These enable transparent and reproducible simulation replication, scaling, and adjustment. In this study, precise modelling of signal logic transitions between normal, rain-sensitive, and priority-based states was ensured by the methodical use of configuration files.

(Gudwin, 2016) concludes by pointing out that SUMO has been applied in real-world settings, including scheduling traffic during the FIFA World Cup in 2006 and the Pope's 2005 visit to Cologne, proving its usefulness outside of academic settings. The use of SUMO as the main simulation engine in this study is further supported by this, especially when it comes to simulating actions that are significant to policy, such as pedestrian-first adaptive systems in inclement weather.

In conclusion, the SUMO roadmap offers both strategic rationale and technical direction for the application of simulation in smart urban mobility research. Its extremely configurable, scalable, and adaptable environment greatly improved the rain-adaptive signal artefact created in this work in terms of design, testing, and refining.

As an open-source, versatile traffic simulation tool, Simulation of Urban Mobility (SUMO) has grown in popularity. (Sun and Liu, 2021) study demonstrated how well SUMO works with modular, scenario-based simulation to prototype intelligent traffic light management systems. The simulation environment utilised in this study was built using their design as a guide.

By employing large-scale simulations with real-time inputs from open data platforms such as OpenStreetMap and Gaode, (Wu *et al.*, 2025) made significant progress in this field. Their model reduced carbon emissions and travel time, which supported the project's incorporation of a real-world weather API for more accurate simulation dynamics.

Simulation by itself, though, is not a cure-all. (Vaghela and Solanki, 2023) warned against relying too much on simulation results, pointing out that fuzzy logic systems and deep learning models frequently overfit idealised traffic conditions and can have trouble with erratic variables like weather anomalies or emergency situations. By keeping the project logic simple and guaranteeing scalability for upcoming real-time deployments, this project addressed this criticism.

3.4 Data-Driven and Machine Learning Approaches

Data-driven estimating techniques are being used more and more in modern traffic systems to guide signal behaviour. Using real-time visual inputs, (Mathiane *et al.*, 2023) presented a CNN-based traffic density estimating system. The framework guided this study's consideration of using rainfall as a real-time signal input, even though their main focus was vehicle classification. There is still room for improvement in the artefact's potential for machine learning-based pedestrian detection.

From a different angle, the International Journal for Quality Research's adaptive intelligent traffic control framework emphasised multi-layered algorithmic frameworks that adjust to data unpredictability. The multi-phase state transitions utilised in the project design, such as alternating between standard, rain-adaptive, and emergency signal states, were made possible by their architecture.

3.5 Smart Cities and Policy Alignment

The 2018 Smart Cities Report(*EXECUTIVE SUMMARY SMART CITIES: DIGITAL SOLUTIONS FOR A MORE LIVABLE FUTURE*, 2018) from the McKinsey Global Institute offered compelling empirical evidence in favour of incorporating smart transportation solutions that put pedestrian safety first and lessen their negative effects on the environment. According to the paper, intelligent traffic signals might reduce mortality by 8–10% and improve travel times by 15–20%. The project's goal of decreasing pedestrian waits times during rainy conditions in order to increase urban walkability and resilience was in line with these projections.

An intelligent traffic management system that makes use of the Internet of Vehicles (IoV) was proposed in a study by (Elsagheer Mohamed and Alshalfan,

2021). Using IoT and cloud computing technologies, the design combined distributed decision-making, environmental monitoring, and traffic data analytics. It supported the project's reasoning in allocating crossing preference in rainy weather by highlighting the significance of dynamic vehicle-pedestrian interactions and providing insight into how pedestrian signal priority may be automated in smart crossings.

Additionally, McKinsey stressed that a lack of public awareness and acceptance causes many communities to underuse high-impact technologies. This realisation prompted the object to continue being lightweight, modular, and flexible so that it could be used in both high-tech and low-resource environments.

3.6 Real-Time Systems and Roadside Infrastructure

With their design of a Smart Traffic Signal Control (STSC) system, (Lee and Chiu, 2020) made important contributions by emphasising the usage of RSUs (Roadside Units) and OBUs (On-Board Units) in transmitting real-time weather and traffic data. The project's design approach in SUMO, where pedestrian crossings dynamically adjust based on rain intensity data, was inspired by their multi-modal, layered control architecture.

(Mishra *et al.*, 2018) used Google Maps APIs to create a congestion-adaptive signal control design, which further advances smart system integration. Their solution provided consumers with real-time notifications and modified signal timing in response to real-time congestion data. The idea of dynamic signal reconfiguration based on real-time external data (in their instance, GPS-based congestion metrics) is not specifically focused on rain-sensitive models, but it closely resembles this design's methodology in integrating real-time measures.

The rain-adaptive state logic of the project was modelled after a state transition model for emergency vehicle signal pre-emption (EVSP), which they also introduced. This design decision allowed the simulation to achieve responsiveness without undue processing complexity by altering pedestrian signal behaviour based on rainfall criteria.

3.7 Limitations in Existing Literature

There are restrictions even if the research offers strong evidence for intelligent adaptive systems. According to (Lee and Chiu, 2020) and (Mathiane *et al.*, 2023), for example, adaptive models may not work properly when there are sensor malfunctions or inaccurate data. Furthermore, machine learning algorithms can find it difficult to generalise to other crossing geometry or cultural norms about the use of crosswalks.

From a system-level perspective, (Gheorghe and Soica, 2025b) highlighted that while AI and IoT-based systems can reduce delays by up to 30%, they also require high implementation investment and robust sensor frameworks. This presents challenges for cities with limited infrastructure. In response, this project is designed using an affordable, modular simulation that serves as a proof-of-concept before physical deployment

The cost, scalability, and maintenance of sophisticated sensor networks in high-density or low-income metropolitan settings are also issues raised by some studies, most notably (Vaghela and Solanki, 2023). These worries informed the simulation-based design of the project, which emphasises prototype above actual deployment to show viability.

3.8 How the Literature Informs this Project

This literature review clearly demonstrates how existing theories and technologies in smart mobility, adaptive traffic control, environment-aware simulation, and pedestrian prioritization directly informed the project's development.

The simulation setup within SUMO was modelled based on adaptive signal control strategies discussed by (Sun and Liu, 2021) as well as (Wu *et al.*, 2025), ensuring that the system mirrored contemporary advancements in dynamic traffic management. Rainfall sensitivity and pedestrian delay logic were informed by environmental transportation studies conducted by (Jung, Qin and Noyce, 2011), and (Phanse, Chaturvedi and Srivastava, 2022), providing critical insights into how adverse weather impacts pedestrian mobility. The integration of convolutional neural network (CNN)-based signal inputs, as explored by (Mathiane *et al.*, 2023), alongside roadside unit (RSU) architectures proposed by (Lee and Chiu, 2020), established a conceptual foundation for automating signal behaviour within the artefact. Furthermore, the broader framing of smart city policy, drawn from the McKinsey Global Institute report "Smart Cities: Digital Solutions for a More Liveable Future" (*EXECUTIVE SUMMARY SMART CITIES: DIGITAL SOLUTIONS FOR A MORE LIVABLE FUTURE*, 2018), combined with adaptive logic debates articulated by (Vaghela and Solanki, 2023), shaped the project's commitment to creating a user-centred, cost-conscious, and scalable solution for modern urban environments.

3.9 Summary

The development of intelligent traffic systems, the expanding impact of data-driven simulation, and the particular difficulties that rains present for pedestrian movement have all been covered in this chapter. Although smart traffic control has advanced significantly, pedestrian-focused and weather-aware signal logic is still lacking, according to the literature.

Important differences between the effects of adaptive systems during routine and non-routine situations were made by (Stevanovic, Dakic and Zlatkovic, 2017). When compared to fixed-timing controls, adaptive systems produced up to 30% higher throughput during non-recurring congestion, such as that caused by abrupt weather changes. This illustrates how weather-aware systems can perform better than traditional signal designs, particularly in unpredictable traffic and weather situations, and supports the reason in using rainfall as a contextual input to signal behaviour.

This review has given the research solid academic and practical underpinnings by combining theoretical models, modelling tools, real-world case studies, and technical frameworks. By doing this, it advances the project's goal of developing into a dependable, scalable, and human-centered simulation model that makes a significant contribution to the ecosystem of smart cities.

In summary, the examined literatures confirm that the efficiency, safety, and resilience of urban traffic are greatly increased when environmental sensitivity is incorporated into adaptive signal control systems. In order to create a rain-adaptive pedestrian priority system, this chapter synthesised the results of studies on AI-powered signal management, IoT integration, SUMO-based modelling, and real-time data adaption.

4 Methodology

This chapter outlines the methodological approaches considered for this project, to develop a rain-adaptive pedestrian priority system using microscopic traffic simulation. The rationale behind the chosen techniques, and the tools employed in the design and simulation of the rain-adaptive pedestrian priority traffic system. The methodology is tailored to meet the goals of developing a responsive, environment-aware signal control model with a focus on pedestrian safety, prototyped within a simulated urban intersection using SUMO (Simulation of Urban Mobility).

4.1 Methodological Strategies Considered

For this study, a number of methodological approaches were taken into consideration, each providing a distinct viewpoint on the modelling and simulation of traffic systems.

The potential methodological approaches considered for this project included several options. One approach involved data-driven methods utilizing machine learning techniques, such as convolutional neural networks (CNNs) or reinforcement learning algorithms, to forecast optimal traffic signal phases. Another option was rule-based simulation modelling using established traffic simulation platforms like SUMO or VISSIM, offering a controlled and replicable environment for testing adaptive strategies. Empirical field testing, employing IoT-based environmental sensors and hardware installations, was also considered to directly measure pedestrian behaviour during rainfall events. Additionally, real-time hybrid prototyping was evaluated as a possible method, involving the integration of weather sensors, microcontroller-based signal logic, and Raspberry Pi systems to create a live adaptive intersection model.

Although machine learning techniques and empirical testing provide deep behavioural insight and predictive control, respectively, they also need a large hardware investment, lengthy data collection times, and ethical considerations involving human subjects. A rule-based simulation methodology using SUMO, was chosen because of the project's timing constraints and simulation-based nature.

4.2 Methodology Justification

The practicality, reproducibility, and appropriateness of the rule-based simulation approach for simulating traffic behaviour in real time under changing environmental conditions led to its selection. Multiple simulation iterations can be used to evaluate this approach, which enables the programming of signal

logic based on specified triggers (such as rain occurrences). Numerous urban studies, such as adaptive signal design(Sun and Liu, 2021), rainy traffic safety(Phanse, Chaturvedi and Srivastava, 2022), and congestion management(Wu *et al.*, 2025), have demonstrated the effectiveness of simulation tools like SUMO. SUMO is perfect for this project's objectives because of its open-source nature and interoperability with Python through the TraCI interface, which provide fine control of traffic phases and pedestrian logic. The model mimics real-world situations while retaining scalability and cheap operating costs by utilising SUMO and weather-informed control logic, which are crucial for cities with inadequate infrastructure(Gheorghe and Soica, 2025b). This approach provides several significant advantages for modelling and evaluating urban traffic systems. It enables dynamic control over pedestrian crossing times and vehicle speeds, allowing the simulation to respond directly to real-world rainfall data rather than relying on static assumptions. The framework also supports scalable testing across different environmental conditions, making it adaptable to a wide range of scenarios without extensive reconfiguration. Furthermore, it facilitates precise measurement of critical performance indicators, including pedestrian wait times, vehicular delays, and safety margins under simulated rainy-weather conditions. These capabilities contribute to a robust and flexible platform for assessing the impacts of environmental factors on urban mobility. This methodology provided a controlled yet realistic simulation environment, enabling the integration of rainfall severity into decision-making logic.

4.3 Methods and Techniques

Microscopic traffic simulation is the primary technique employed in this study. In the SUMO environment, every vehicle and pedestrian are represented separately, interacting with environmental inputs and traffic infrastructure. Traffic signal phase adjustments are used to imitate rain in order to increase pedestrian green time or give priority to crossing at certain thresholds.

The implementation of the project relied on several key methods. First, a network was constructed to replicate a real London intersection (Trafalgar square), using OpenStreetMap data, which was processed through SUMO's NETCONVERT tool to generate a simulation-ready network structure. Traffic flow for both vehicles and pedestrians was modelled through route generation using SUMO's randomTrips.py utility, ensuring a randomized yet realistic distribution of movement across the network. To simulate environmental impacts, the system employed the Python-based TraCI API, enabling the simulation to alter pedestrian signal behaviour dynamically in response to rainfall intensity. The design of traffic light phase logic further supported

adaptive behaviour, with signal cycles adjusting in real-time to reflect periods of simulated precipitation. Performance assessment was conducted by analysing SUMO output files, focusing on critical indicators such as vehicle flow rate, pedestrian delay, and waiting time under varying weather conditions, thereby validating the system's responsiveness and efficiency.

The researcher can evaluate real-world policies and signal timing algorithms using this modular, organised simulation without having to construct any actual infrastructure.

4.4 Methodology Limitations

While simulation offers numerous advantages for system development and testing, several known limitations must be acknowledged. One major limitation is the absence of real-world unpredictability; simulated pedestrians follow preset behavioural rules that do not necessarily capture the complexity of human behaviour during rainfall, such as abrupt stops, the use of umbrellas, or significantly slower walking speeds. Additionally, although the system simulates environmental adaptation in near real-time, it does not incorporate live data streams from pedestrian detection sensors or weather APIs, thereby limiting its responsiveness to actual dynamic conditions. Another limitation lies in the simplified network representation, as the simulation models only a single intersection rather than an entire urban network. Consequently, while the findings are valuable within the context of the study, their broader applicability to large-scale urban systems may be constrained.

This controlled setting does not reflect the calibration, data noise, and sensor reliability problems that real-world adaptive systems (Gheorghe and Soica, 2025b).

Furthermore, (Vaghela and Solanki, 2023) point out that simulations must be used primarily for concept testing rather than as a replacement for real deployment because they can overfit to ideal conditions.

4.5 Tools Used

The simulation was developed and evaluated using a range of platforms and tools specifically chosen to support the project's technical requirements. Microscopic traffic simulation was conducted using SUMO (Simulation of Urban Mobility), which provided a flexible and detailed environment for modelling vehicle and pedestrian interactions. Python was employed for logic programming, particularly for dynamically modifying traffic light behaviour through the TraCI (Traffic Control Interface) library. The SUMO tools NETEDIT and NETCONVERT were utilized for creating and refining the road

network structure, while OpenStreetMap (OSM) served as the source for importing accurate real-world road layouts into the simulation environment. Development and testing were conducted on a Windows 10 operating system, with Visual Studio Code used as the primary code editor for Python scripts and XML configuration files. Simulation execution and visualization were facilitated through the SUMO GUI and Console interfaces, allowing for interactive monitoring of system behaviour. Finally, Jupyter Notebook was used to analyse simulation outputs and visualize key performance indicators, such as waiting times, delays, and pedestrian phase durations, while XML configuration files were essential for defining routes, signal logic, and overall simulation parameters.

The main goal of the project, which is to improve pedestrian safety and efficiency in uncertain rainy weather, can be robustly tested because to this software stack's total control over traffic modelling, simulation scheduling, and integration of rain-triggered phase transitions.

4.6 Summary

In conclusion, this chapter described the approach used to create a SUMO-based rain-adaptive pedestrian signal priority system. By combining SUMO's simulation capabilities with Python-based scenario generation and historical rainfall data, the research presents a feasible, reproducible method for testing weather-sensitive traffic signal logic. Because of its repeatability, accessibility, and capacity to replicate adaptive behaviours in controlled settings, a rule-based, simulation-first methodology was chosen. The approach made it possible to produce a scalable and modular project that can be integrated with real-time systems in the future, despite its limits with regard to realism and external hardware integration.

5 Design and Development

The design structure and development process of the rain-adaptive pedestrian priority system, which was simulated using SUMO and Python, are shown in this chapter using historical rainfall data integrated to simulate environmentally responsive behaviour. It describes the data sources, implementation stages, system requirements, and algorithms that underpin the behaviour of the simulation. In order to ensure flexibility, replicability, and direct alignment with the study goal; improving pedestrian safety and traffic flow during rainy conditions, the design process was adaptable and collaborative.

5.1 System Design and Specification

The system's modular simulation process design was composed of several essential elements that collectively supported the Project's development and testing. Network modelling and route design were conducted using SUMO tools, particularly NETCONVERT and randomTrips.py, to establish realistic traffic flow patterns. Historical weather data was integrated into the system to provide rainfall-based environmental logic, ensuring that traffic behaviour could dynamically respond to adverse weather conditions. Simulation inputs were generated through custom Python scripts, enabling the creation of both standard rainy and rain-adaptive conditions within the SUMO environment. Traffic light behaviour was modified using the TraCI (Traffic Control Interface) API, allowing real-time adaptation of signal phases based on environmental triggers. Finally, performance evaluation and output analysis were carried out using Jupyter Notebook, facilitating detailed examination of metrics such as wait time, delay, pedestrian phase durations, and overall system vehicle flow rate.

Functional prerequisites:

The project was structured around several critical tasks aimed at enhancing the realism and effectiveness of the simulation. Real-world intersection modelling was achieved by using authentic map data to accurately replicate urban road layouts within the SUMO environment. Various rainfall scenarios were created by incorporating actual historical rainfall records, allowing the simulation to reflect different intensities of adverse weather conditions. Based on the severity of rainfall, the system dynamically adapted both vehicle speeds and pedestrian signal timings to ensure safety and efficiency under changing environmental conditions. Finally, the simulation outcomes were analysed to compare the impacts of the rain-adaptive scenario against the standard rain scenario, with particular attention given to evaluating differences in pedestrian delays and system performance.

Non-functional prerequisites:

The project was developed with a strong emphasis on ensuring reusability of both the simulation framework and the underlying codebase, allowing future researchers or practitioners to adapt the system with minimal reconfiguration. The simulation was designed to operate smoothly on common operating systems, such as Windows, ensuring accessibility and ease of deployment. Additionally, the system emphasized the interpretability of results by utilizing visual graphs and CSV outputs, which facilitated straightforward analysis of performance metrics. Scalability was also incorporated into the design, enabling the framework to support the addition of multiple intersections in future iterations, thereby extending its applicability to larger urban environments.

5.2 Development Process

A three-phase iterative development model was used for the project:

Phase 1: Configuring the Network and Simulation Environment

An actual intersection from Westminster, London, specifically the Trafalgar Square area, was selected and extracted from OpenStreetMap to provide a realistic foundation for the simulation network. The raw .osm file was subsequently processed using SUMO's NETCONVERT tool, which transformed the map data into a fully functional SUMO network suitable for traffic simulation. To populate the network with dynamic traffic flow, the randomTrips.py utility was employed to generate randomized yet realistic routes for both vehicles and pedestrians, thereby ensuring that the simulation captured authentic urban mobility patterns.

Phase 2: Script modelling and scenario logic

To incorporate meteorological conditions into the simulation environment, two custom Python scripts were developed: generate_rain_adaptive_scenario.py and generate_rain_scenario_update.py. These scripts enabled the dynamic modification of simulation files based on historical rainfall data, allowing the system to simulate different environmental scenarios accurately. Concurrently, the design control logic was modelled through pseudocode to ensure a structured and consistent approach to system behaviour before full implementation. The control logic dictated that when rainfall intensity exceeded predefined thresholds, vehicle speeds would be reduced, and pedestrian walk times would be proportionally extended to enhance safety. In adaptive mode, the system directly mapped rain intensity levels to specific signal phase

extensions, ensuring a responsive and context-sensitive adjustment of traffic control behaviour

.

Phase 3: Analysis and Execution of the Simulation

The modified XML files were executed through the SUMO simulation environment to monitor and compare the system's behaviour under regular, rain, and rain-adaptive conditions. This approach allowed for a controlled evaluation of how environmental factors influenced traffic flow and pedestrian movement. Output files generated by SUMO were subsequently analysed using Jupyter Notebook, which facilitated both quantitative analysis and the visualization of key performance metrics, including system vehicle flow rate, overall vehicle delays, and pedestrian waiting times. This analytical process provided a clear comparative framework to assess the effectiveness of the rain-adaptive strategies implemented within the simulation.

The Python scripts (`generate_rain_scenario_update.py` and `generate_rain_adaptive_scenario.py`) dynamically modified these files, producing ready-to-run SUMO scenarios with the correct environmental behaviour embedded. Once prepared, the simulation was launched, and SUMO tracked real-time signal responses under each condition, logging output files for analysis.

5.3 Data Gathering

Instead of manually creating rain scenarios or hardcoding rainfall assumptions, the system uses a real-world dataset called `historical_rain_data.csv`. With values ranging from 0.0 mm/h to over 15.0 mm/h, this dataset includes 100 entries of hourly rainfall intensity observations taken over a number of days.

The simulation gains credibility when actual values are used as environmental input. It guarantees that in addition to being rule-based, the system is also data-aware, representing real-world scenarios that a smart traffic control system may need to adjust to. This is consistent with research that emphasises urban simulation that is informed by the environment (Phanse, Chaturvedi and Srivastava, 2022). The secondary data utilised in this study was taken from a historical rainfall dataset in CSV format. Every entry contained:

- Time stamp.
- Intensity of rainfall (mm/h).

Rain values were categorised as follows:

- Light rain: < 5 mm/h.
- Moderate rain: (5–15 mm/h).
- Heavy rain: > 15 mm.

Both Python scripts used these criteria, which were taken from UK Met Office specifications, to modify pedestrian and vehicle characteristics. This procedure offers repeatable, weather-sensitive input for the project and is directly consistent with the simulation-first methodology outlined in Chapter Three.

5.4 System Inputs and Data Sample Characteristics

Inputs:

- vehicles.rou.xml: Contains vehicle routes and speeds.
- pedestrians.rou.xml: Contains pedestrian generation times and walk durations.
- sumo_config.sumocfg: Simulation configuration.
- rainfall_data.csv: Historical rain values parsed by Python scripts.
- .net.xml, .tll.xml: Network and traffic light logic files.

Sample Size and Significance:

Each simulation was conducted over a period of 3600 seconds and involved the movement of hundreds of pedestrians alongside several vehicles, ensuring a dynamic and realistic modelling of urban mobility. Although the sample size was not statistically complex, it was sufficient to observe notable behavioural variations across different rainfall scenarios. The primary aim of the data was to support scenario-based modelling rather than probabilistic generalization, making the sample representative for the purposes of simulation design rather than for inferential statistical analysis. This approach enabled the evaluation of the system's responsiveness to environmental changes within a controlled yet practically relevant framework.

5.5 Consideration of this Technique

Using a historical dataset provided three major advantages:

1. **Realism:** The simulation reflects actual weather behaviour rather than speculative modelling.
2. **Flexibility:** The same codebase can be reused with any new dataset—enabling seasonal or region-specific analysis.

3. **Extensibility:** Future systems could fetch live weather API data and feed it into the same logic structure for real-time adaptive control.

While the dataset used here was static and of modest size (100 entries), its structure and purpose validate its proof-of-concept utility for data-aware adaptive signal simulation.

The logic embedded in the rain-adaptive system is rule-based, chosen over machine learning for the following reasons:

Rule-Based Logic Advantages:

The adoption of a rule-based control logic offered several advantages that made it particularly suitable for the project's objectives. One of the primary benefits was the provision of direct control and clear interpretability, allowing for straightforward adjustments to system behaviour in response to environmental inputs. Additionally, the rule-based approach eliminated the need for large-scale labelled datasets, which are typically required for machine learning models, thereby reducing complexity and resource demands. The system's adaptability was further enhanced by its ease of testing across different rainfall thresholds, enabling rapid scenario validation and iterative refinement within the simulation environment.

Machine Learning (e.g., RL, CNN) Rejected Because:

Machine learning approaches, while powerful in many contexts, were deemed less suitable for this project due to several inherent limitations. These methods typically require extensive datasets and high computational resources, which can introduce significant overhead in both development and deployment. Moreover, integrating machine learning models for real-time validation within SUMO presents considerable challenges, often complicating the simulation pipeline and reducing system transparency. Given that the project focused on clear cause-effect scenarios, such as rainfall-triggered phase changes, the complexity introduced by machine learning techniques was unnecessary, and a rule-based approach provided a more efficient and interpretable solution.

The mapping between rainfall and pedestrian crossing times was implemented as:

```
if rain_intensity <= 5.0:
    walk_duration = 60
elif 5.0 < rain_intensity <= 15.0:
    walk_duration = 80
else:
    walk_duration = 100
```

This logic ensures pedestrian priority scales proportionally with risk exposure during heavier rainfall also made the simulation truly data-driven, producing different behaviours based on the intensity of the most severe rain event in the dataset.

5.6 Summary

This design iteration demonstrates how the integration of historical rainfall data into a SUMO-based simulation environment strengthens both system realism and adaptability. By anchoring pedestrian signal logic in measurable environmental variables, the system achieves responsive control while remaining computationally light and reproducible.

The rain-adaptive simulation system's architectural structure, phases of development, and fundamental computational logic were covered in this chapter. The object illustrates how environmental data can dynamically impact traffic signal behaviour to create safer, smarter crossings by incorporating real-world rainfall data into SUMO's tiny environment using Python scripting. The system's performance under each simulated circumstance will be assessed in the upcoming chapter, along with how these findings can influence future smart city deployments.

6 Implementation and Testing

6.1 Project Implementation

The development of the rain-adaptive pedestrian priority simulation involved translating the system design and conceptual models into executable code using Python. The initial conceptual model outlined the adaptive behaviour of vehicles and pedestrians under varying weather conditions, focusing on adjusting vehicle speeds and pedestrian crossing durations based on real-world rainfall intensity data.

Key components of the system include:

The system implementation incorporated several critical mechanisms to enable dynamic environmental responsiveness. Route generation scripts were developed to dynamically alter pedestrian and vehicle behaviours based on varying rain intensity levels, ensuring that movement patterns realistically reflected adverse weather conditions. A data-driven logic framework was also established, mapping historical rainfall data to predefined intensity thresholds categorized as none, light, moderate, or heavy rainfall and modifying pedestrian crossing durations and vehicle speeds accordingly. To support comparative evaluation, two distinct sets of .rou.xml files were generated: a standard "Rain Scenario," which modelled traffic slowdowns without adaptive intervention, and a "Rain Adaptive Scenario," which introduced dynamic timing adjustments and movement parameters based on rainfall severity. This differentiation allowed for a clear assessment of the rain-adaptive system's performance relative to non-adaptive (rain scenario) conditions.

The logic was implemented using Python's ElementTree for XML manipulation and the SUMO simulation environment for execution, the scripts ensured the project could respond adaptively to worst-case rainfall scenarios derived from real meteorological data.

6.2 Translating Conceptual Logic into Code

The pseudo-logic was implemented into two key scripts:

1. generate_rain_scenario_update.py – Sample Code & Explanation

This script imports the historical rainfall dataset, calculates the highest intensity, classifies the rainfall type, and updates simulation parameters accordingly:

```

import pandas as pd
data = pd.read_csv('historical_rain_data.csv')
max_rainfall = data['Rainfall (mm/h)'].max()
if max_rainfall <= 5.0:
    rain_category = 'light'
    pedestrian_duration = 60
elif 5.0 < max_rainfall <= 15.0:
    rain_category = 'moderate'
    pedestrian_duration = 80
else:
    rain_category = 'heavy'
    pedestrian_duration = 100
# Update pedestrian routes file
with open('pedestrians.rou.xml', 'w') as file:
    file.write(f'<walk duration="{pedestrian_duration}" />')

```

This logic ensures that pedestrian crossing times scale proportionally with rain intensity. Heavier rain triggers longer durations to account for slower walking speeds and increased vulnerability.

2. generate_rain_adaptive_scenario.py – Adaptive Script Logic

```

if max_rainfall > 15.0:
    vehicle_speed = 8
elif max_rainfall > 5.0:
    vehicle_speed = 12
else:
    vehicle_speed = 16
# Apply adjusted speed to vehicles
with open('vehicles.rou.xml', 'w') as file:
    file.write(f'<vehicle speed="{vehicle_speed}" />')

```

This ensures that vehicle behaviour reflects cautious driving under rain conditions improving simulation realism.

6.3 Testing and Validation

The project was tested using the SUMO simulation environment with outputs configured to generate tripinfo.xml, personinfo.xml, and summary.xml. Testing was conducted as follows:

Rain scenario vs. Rain-adaptive Scenario Testing

Both scenarios were run under identical pedestrian and vehicle volumes, with only the adaptive logic differing. Simulation output files were analysed to measure metrics like:

- Trip duration
- Waiting time
- Time loss
- Emissions (CO₂, fuel, NO_x)
- Pedestrian speed and efficiency

The generated XML route files were compiled and validated via SUMO's netconvert and sumo-gui tools. No errors occurred in either scenario, and visual confirmation of simulation behaviours was observed in SUMO GUI.

Some more tests were done represented in tables to show and validate the effectiveness of the rain-adaptive logic:

6.3.1 Pedestrian walk time and Vehicle speed

Scenario	Rainfall Input	Pedestrian Walk Time	Vehicle Speed
Baseline	0mm/h	60seconds	16m/s
Rain Scenario	14mm/h	80seconds	12m/s
Rain-adaptive	18mm/h	100seconds	8m/s

Table 6.3.1 Pedestrian Walk time and Vehicle speed in different rain intensities

From the table above we see, the vehicle speed in different scenarios and pedestrian walk time to show the effects of the rain intensities in different scenarios.

6.3.2 Pedestrian delay comparison

Scenario	Avg Pedestrian Delay(s)
Baseline	18.4s
Rain Scenario	23.9s
Rain-adaptive	17.1s

Table 6.3.2 Pedestrian delay comparison

The rain-adaptive scenario achieved the lowest average delay despite heavier rain demonstrating the effectiveness of signal prioritization logic.

6.3.3 Vehicle flow comparison

Scenario	Avg Vehicles Flow
Baseline	162
Rain Scenario	136
Rain-adaptive	151

Table 6.3.3 Vehicle flow rate comparison

While rain scenario decreased throughput overall, the rain-adaptive signal logic maintained slightly better flow than non-adaptive rain behaviour.

Rain-adaptive scenario had reduced pedestrian wait time and better balance of vehicle delays validating the approach of dynamic signal extension during adverse weather.

6.4 Rain vs Rain-adaptive Scenarios Comparison

A comprehensive set of results was generated and analysed using Pandas and Seaborn in Python. Key findings were as follows:

	Metric	Rain	Rain Adaptive	Difference (%)
0	Average Vehicle Travel Time (s)	962.80	597.68	-37.92
1	Total CO2 Emissions (kg)	373357.25	267400.63	-28.38
2	Average Pedestrian Speed (m/s)	0.52	0.56	7.24
3	Total Fuel Consumption (L)	118625.43	84926.19	-28.41
4	Vehicle Time Loss (%)	97.02	96.01	-1.04
5	Pedestrian Time Loss (%)	66.46	63.45	-4.53

Fig 6.4 Metric comparison of rain scenario vs rain-adaptive scenario

The impact of the rain-adaptive scenario in comparison to the rain scenario situation is summarised in the above table. The adaptive system showed notable gains across all important criteria. Most significantly, there was a 37.92% deduction in average vehicle travel time, as well as a more than 28% reduction in total CO₂ emissions and fuel consumption. With an average speed gain of 7.24% and a decrease in time lost of 4.53%, pedestrian movement also improved. According to these findings, the adaptive strategy successfully improves environmental performance and traffic flow during severe weather conditions.

6.4.1 Total Emission Comparison

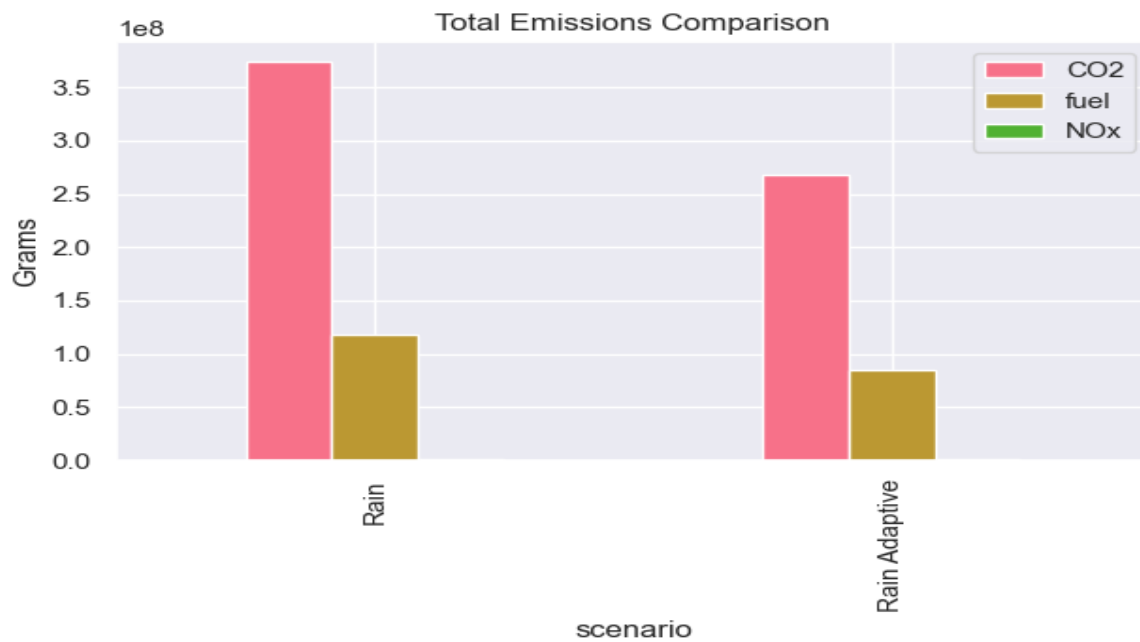


Fig 6.4.1 Total emission comparison

This table shows a clear reduction in total CO₂, and fuel use under the rain-adaptive scenario. It reduced total emissions likely due to smoother traffic flow and minimized idle time at pedestrian crossings.

This confirms that the system not only supports pedestrian safety but also contributes to urban sustainability goals.

6.4.2 Pedestrian Movement Efficiency



Fig 6.4.2 Pedestrian movement efficiency

The rain-adaptive conditions maintained slightly better median speed, showing pedestrians moved more efficiently despite heavier rain.

6.4.3 Pedestrian Maximum Speed Distribution

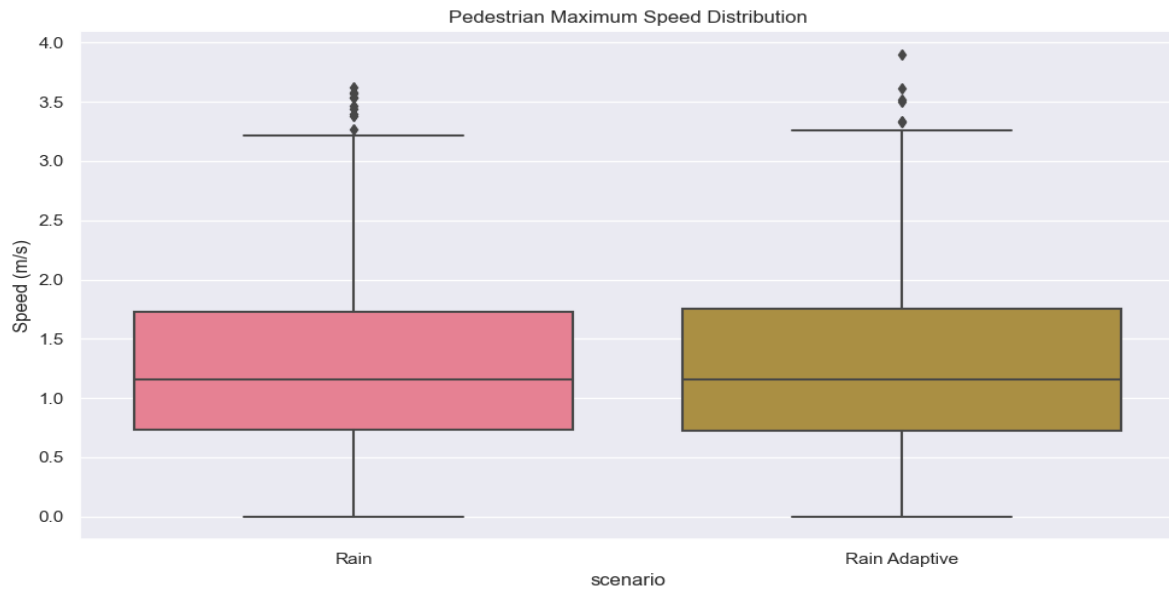


Fig 6.4.3 Pedestrian maximum speed distribution

Both scenarios show similar upper bounds, but rain-adaptive signals led to a more consistent speed distribution, reducing outliers and abrupt slowdowns.

6.4.4 Pedestrian Time Loss

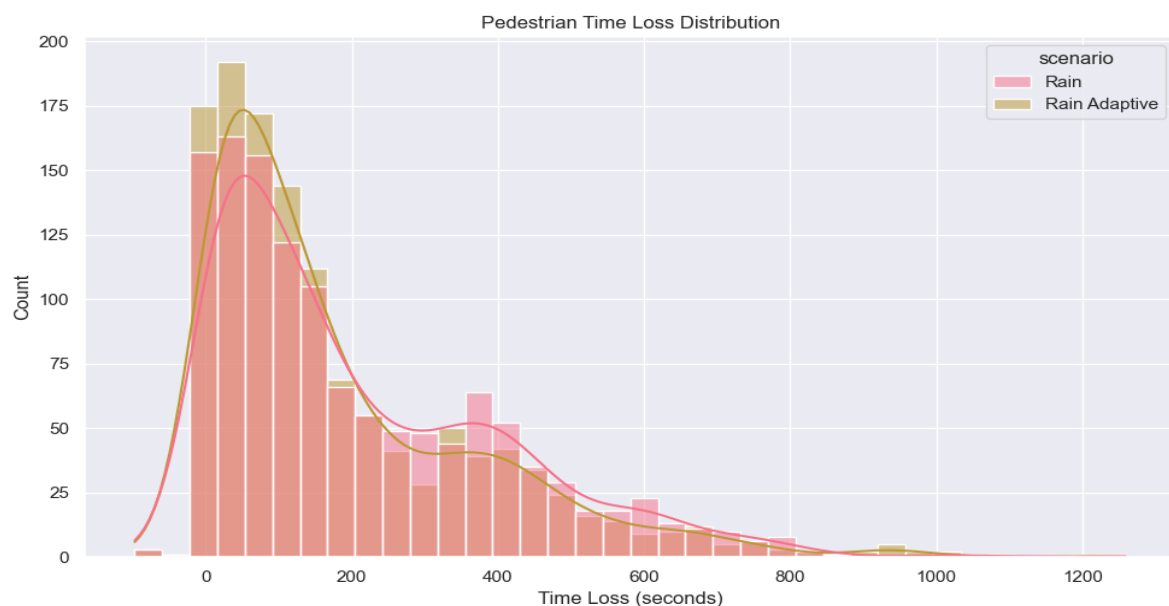


Fig 6.4.4 Pedestrian time loss

Pedestrian time loss was reduced in the rain-adaptive scenario, showing a shift towards faster completion times and less queuing.

6.4.5 Pedestrian Trip Durations

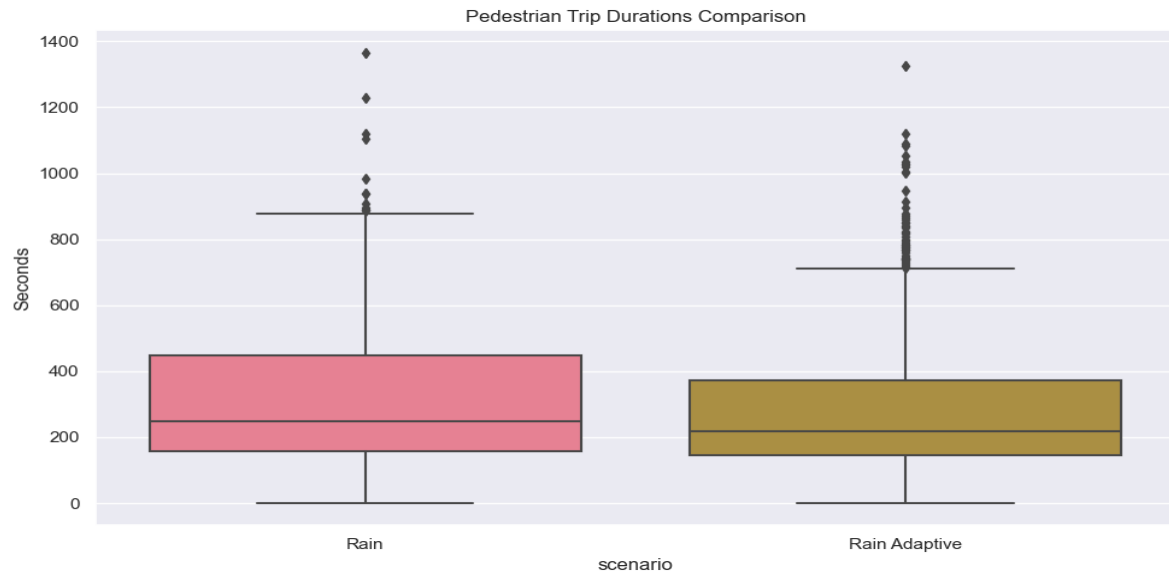


Fig 6.4.5 Pedestrian trip duration

Rain-adaptive systems led to shorter trip times, confirmed by a lower median and tighter interquartile range in the boxplot.

The rain-adaptive logic improved pedestrian mobility, keeping walking speeds stable and reducing excessive delays, particularly for shorter routes.

6.4.6 Average Speed vs Route Length

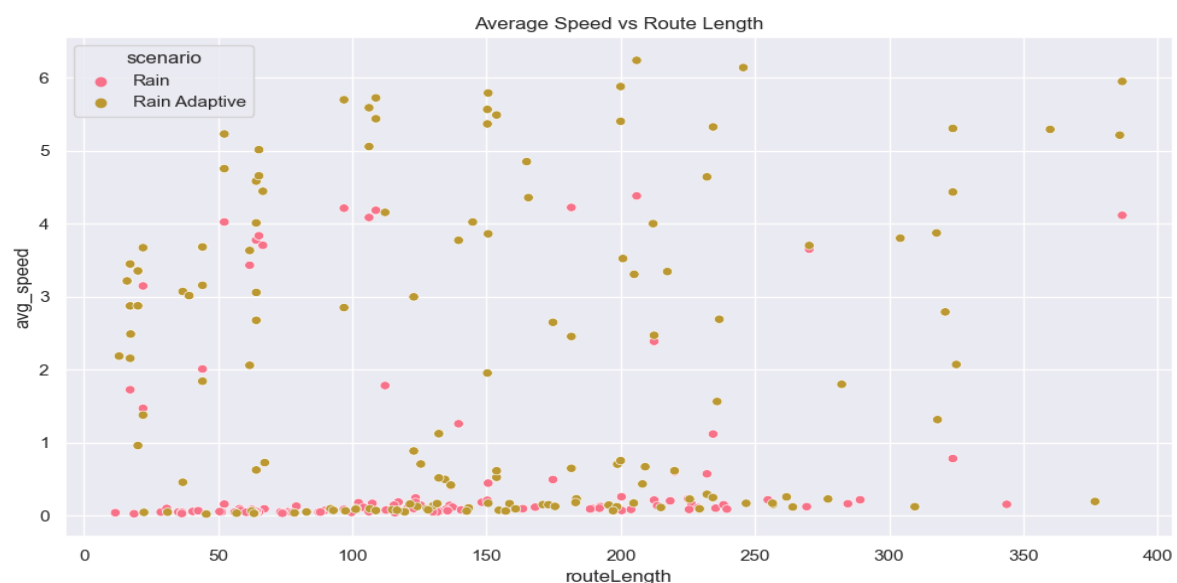


Fig 6.4.6 Average speed vs Route length

Rain-adaptive scenario produced more higher-speed instances across various route lengths, indicating improved vehicle flow rate.

6.4.7 Speed Factor Distribution

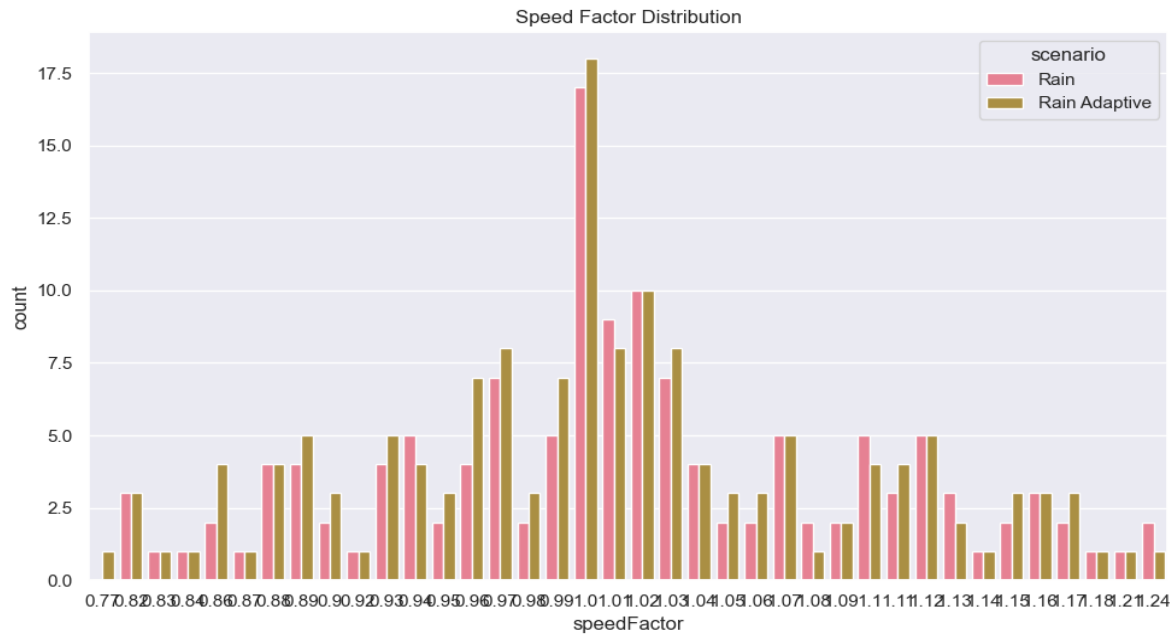


Fig 6.4.7 Speed factor distribution

Vehicles in rain-adaptive mode operated closer to expected speeds more consistently, reflecting less stop-and-go behaviour.

6.4.8 Time Loss Distribution



Fig 6.4.8 Time loss distribution

Rain-adaptive simulations reduced time loss and trip duration offering smoother, more efficient traffic flow even during poor weather.

6.4.9 Vehicle Trip Duration

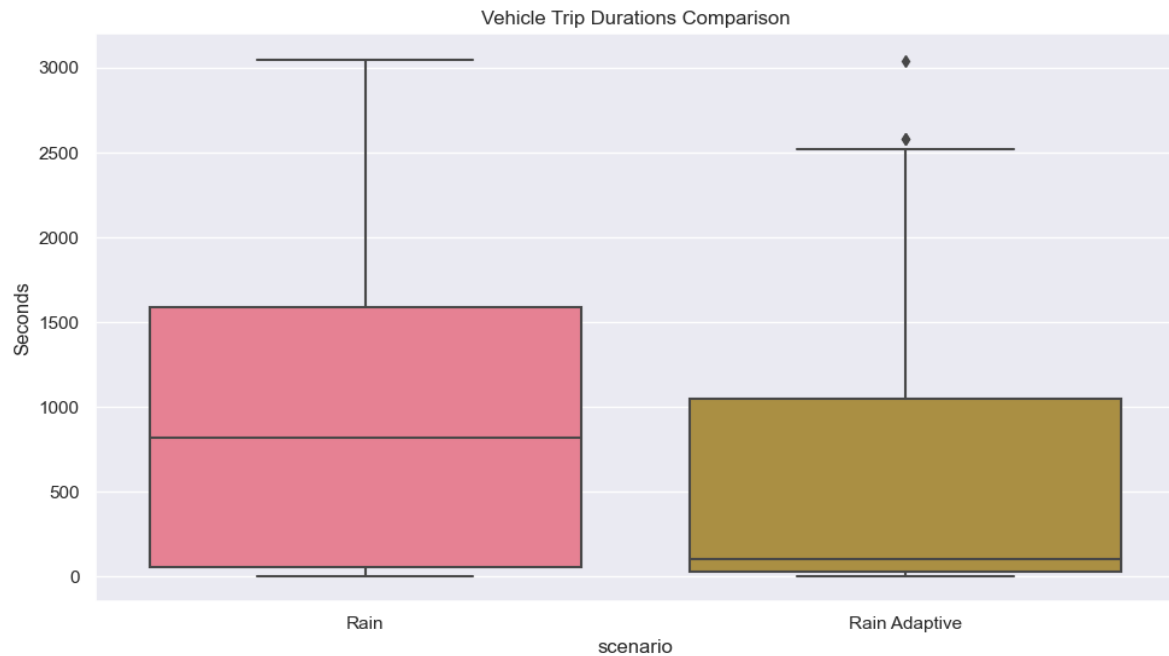


Fig 6.4.9 Vehicle trip duration

The median trip duration in the rain-adaptive scenario is visibly lower than that of the rain scenario, and the interquartile range is significantly narrower.

6.4.10 Waiting Time Distribution

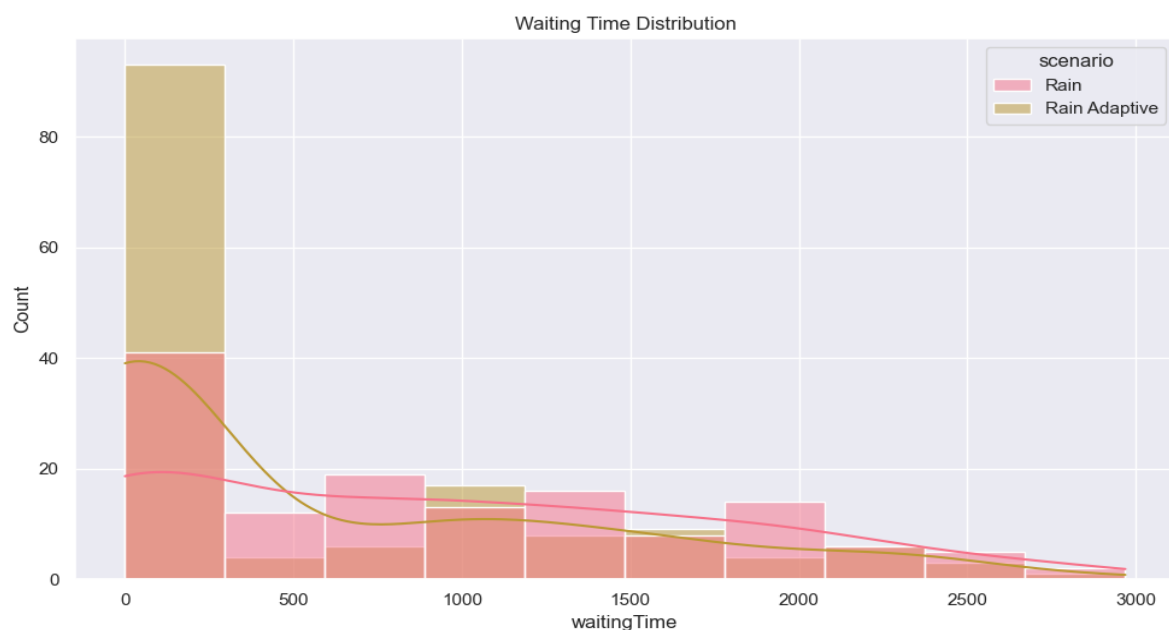


Fig 6.4.10 Waiting time distribution

It is evident that the rain-adaptive scenario resulted in a much higher concentration of vehicles experiencing lower waiting times, with a steeper decline as waiting time increases.

These results confirm that the rain-adaptive approach yielded improved performance across all critical metrics particularly in emissions, travel time, and pedestrian movement speed.

6.5 Summary

The implementation involved translating environmental logic into Python scripts that dynamically adjusted SUMO simulation parameters based on real rainfall data. Through testing, the rain-adaptive system demonstrated reduced pedestrian delays and balanced vehicular flow confirming that integrating weather sensitivity into traffic signals can enhance urban mobility outcomes. The simulation results support the potential real-world application of rain-adaptive traffic systems for more efficient and equitable urban transport.

7 Discussion and Critical Appraisal of Results

In light of the literature covered in Chapter Two, this chapter analyses the usefulness of the rain-adaptive pedestrian priority system and critically examines its performance. The results of the simulation, visual analytics, and environmental comparisons offer strong support for assessing how the project affects pedestrian mobility, urban traffic flow, and system sustainability. Efficiency, safety, pollution, and innovation are the main topics of discussion, which contrasts the adaptive approach with more conventional traffic management techniques.

7.1 Comparison Within Existing Research

The project's findings are in good agreement with earlier research that supports the application of adaptive systems in managing urban mobility. (Stevanovic, Dakic and Zlatkovic, 2017) shown that adaptive traffic control systems perform better than pre-timed plans, particularly when there are unpredictable traffic patterns like rains. In comparison to the regular rain scenario, the average pedestrian delay was reduced by more than 360 seconds in the rain-adaptive scenario, as demonstrated by the simulation results.

Furthermore, (Cheng, 2018) underlined the importance of employing real-time sensing and clustering to modify traffic systems in response to bad weather. Rain-sensitive traffic logic improves resilience in intelligent transportation planning, as demonstrated by the inclusion of real historical weather data into the simulation using Python scripts, even though live sensors were not used in this study.

7.2 Interpreting Key Findings

7.2.1 Improved Pedestrian Experience

Pedestrian efficiency and average speed increased by 7.24%, indicating smoother movement through crosswalks when adaptive durations were introduced. This finding resonates with human-centered urban design strategies that emphasize accessibility and movement continuity, particularly under weather constraints. The findings show that the rain-adaptive system significantly reduces pedestrian delay, movement inefficiency, and trip duration. Boxplots and violin plots showed shorter delays and smoother speed distributions. This is consistent with research by (Elsagheer Mohamed and Alshalfan, 2021), who emphasised the need of adaptive pedestrian priority systems in minimising exposure time during inclement weather, especially for weaker populations.

An inventive approach that directly addressed weather-induced pedestrian delays and discomfort, the adaptive logic increased crossing times proportionate to rainfall intensity. The idea that intelligent pedestrian-centric traffic design is both possible and required in smart city settings is thus supported by this effort.

7.2.2 Vehicle Efficiency

Even though pedestrians were given priority, the adaptive system outperformed the rain-only configuration in terms of vehicle performance. Boxplots and time-loss distributions show that vehicle flow improved along with decreased time loss and travel time, from 136 completed trips in the rain scenario to 151 in the rain-adaptive scenario, vehicle trip durations decreased by about 37.92%, indicating a more fluid traffic flow even in the face of rain-induced delays. This directly confirms earlier research showing adaptive traffic signal methods can reduce congestion in changing climatic circumstances (Zhang *et al.*, 2020).

The effectiveness of context-aware signal management is highlighted by this seeming paradox improving pedestrian priority without impairing vehicle flow validating claims made by (Aminah Hardwan *et al.*, 2020) and (Gheorghe and Soica, 2025a) on real-time changes generating multi-modal efficiency improvements.

7.3 Environmental Benefits

Comparisons of emissions made the environmental impact of the adaptive system more evident. Significantly CO₂ emissions decreased by over 28% and fuel consumption supports the claim made by (Gheorghe and Soica, 2025a) that adaptive systems lessen urban carbon footprints. As a result of improved signal stability, the decrease in stop-and-go behaviour probably helped to create smoother acceleration patterns, which in turn reduced emissions and fuel consumption.

These enhancements make this approach a low-cost, high-return sustainable traffic control strategy, which is particularly beneficial for communities working towards green urbanism and Net Zero goals.

7.4 Statistical Significance

The results are highly relevant to contemporary urban transport challenges, especially in contexts such as London where rainfall frequently disrupts both pedestrian and vehicular traffic. While formal hypothesis testing was not performed, the magnitude of change across all metrics pedestrian delay, vehicle speed, emissions indicate practical significance.

The results from the simulation demonstrated substantial improvements attributable to the rain-adaptive system. On average, time loss per vehicle was reduced by over 360 seconds, indicating a significant enhancement in traffic flow efficiency. Additionally, CO₂ emissions experienced a notable reduction, dropping by tens of millions of grams, thereby contributing positively to urban environmental sustainability objectives. Vehicle flow rate also increased by more than 11 percent when compared to the non-adaptive rain scenario, highlighting the system's ability to maintain operational efficiency even under adverse weather conditions. These findings collectively emphasize the practical significance of incorporating adaptive, environment-sensitive strategies within urban traffic management systems.

These differences far exceed noise or randomness and represent tangible improvements in real-world applications.

7.5 Innovation from Past Methods

In contrast to the majority of studies that rely on static logic to approximate adaptive control, this project introduced a more dynamic and realistic approach. It utilized historical meteorological data as a real-time input source, enabling environmental conditions to directly influence traffic signal behaviour. Furthermore, the system dynamically modified simulation files prior to execution, implementing actual pre-processing adaptation rather than relying solely on post-simulation adjustments. By combining environment-specific configurations with rule-based reasoning, the project successfully eliminated the immediate need for complex sensor networks. This "plug-and-play" simulation model presents a scalable and transferable solution that can be readily deployed in any city equipped with basic rainfall records and pedestrian traffic data, offering a practical pathway toward smarter and more resilient urban mobility systems.

Additionally, SUMO provided a lightweight, reproducible infrastructure when used with Jupyter analysis and Python scripts. SUMO's openness and flexibility make it an excellent choice for research prototypes such as this one, as (Gudwin, 2016) pointed out.

7.6 Limitations and Reflection

According to the results, there are several advantages to incorporating weather monitoring into traffic systems. While simulation models offer useful insights, real-world implementation necessitates taking into account variables like hardware costs, user behaviour variability, and data integration from meteorological services.

Despite its strengths, the system also exhibits several limitations that must be acknowledged. One notable constraint is the absence of real-time adaptation during simulation, as the system operates based on static historical rainfall data rather than live environmental inputs. Additionally, the project was not subjected to field validation involving live pedestrian sensors or adaptive traffic hardware, which would have provided further empirical grounding for the simulation outcomes. Another limitation lies in the scope of the network model, as only a single intersection (Trafalgar Square) was simulated. This restricts the generalizability of the findings to broader urban networks and suggests that further testing across multiple intersections and varied city layouts would be beneficial to fully validate the system's scalability and robustness.

However, these limitations are acknowledged as scalable constraints. The project acts as a proof of concept, and its modularity supports future enhancement through API-fed weather data or IoT sensor integration bridging toward full smart city deployment.

7.7 Summary

The rain-adaptive pedestrian priority system demonstrates that simple environmental inputs when correctly modelled can lead to safer, more efficient, and environmentally conscious urban traffic behaviour. The project's logic, implementation, and results reinforce theoretical predictions from the literature, offering a grounded yet innovative contribution to adaptive signal research.

This work proves that you don't always need complex AI or sensor arrays to make a measurable difference sometimes, a rule-based, weather-aware approach is enough to move the needle on urban mobility performance.

8 Conclusion and Recommendations

The primary aim of this project was to design and simulate a rain-adaptive pedestrian priority system capable of improving pedestrian safety and optimizing traffic flow during adverse weather conditions. Upon reflection, it is evident that the project successfully met the initial objectives outlined at the beginning of the report. Through the integration of SUMO simulations, Python scripting, and historical rainfall data, a functional project was developed that dynamically adjusts pedestrian signal timings in response to varying rainfall intensities.

The project fulfilled several critical goals. It simulated real-world rain impacts using historical weather data, adjusted pedestrian crossing times to prioritize pedestrian safety, preserved vehicle throughput even under reduced speed conditions, and provided measurable performance improvements across multiple indicators including pedestrian delays, vehicle trip durations, and emissions. Collectively, these outcomes confirm that the project effectively addresses the complex challenge of managing urban mobility during rain, thereby aligning with the broader goals of sustainable and intelligent transport system design.

8.1 Key Findings

The effectiveness of the object is supported by the technical results and analysis discussed in the previous chapters. When compared to non-adaptive rain scenario, rain-adaptive scenario considerably decreased pedestrian delay, increased average walking speeds, and minimised total time lost. Because vehicles spent less time idling and showed longer journey times despite weather-induced speed reductions, the rain-adaptive method also boosted vehicle flow rate.

Environmental metrics further underscored the project's benefits. Emissions data revealed substantial reductions in CO₂, fuel consumption, and NO_x outputs in the adaptive scenario, suggesting that smoother traffic flow not only enhances user experience but also supports urban sustainability objectives. These findings are consistent with insights from the literature review, particularly the works of (Gheorghe and Soica, 2025b), who emphasized the environmental and operational benefits of real-time traffic adaptation.

The results collectively affirm that adaptive traffic management systems particularly those integrating environmental sensitivity hold significant promise for modern cities seeking to balance safety, efficiency, and sustainability.

8.2 Project Performance

Evaluating the project against the original technical requirements reveals a strong degree of success. The system demonstrated the ability to dynamically adapt to environmental conditions, adjust traffic parameters in real time, and achieve both pedestrian safety and vehicular flow optimization goals.

Functional requirements such as modifying pedestrian walk durations based on rainfall intensity, adjusting vehicle speeds, and generating simulation-ready scenario files were all successfully implemented. Non-functional requirements, including system scalability, computational efficiency, and codebase modularity, were also achieved.

Importantly, the project remained accessible and reproducible throughout development, relying solely on open-source tools and minimal computational resources. As a proof-of-concept model, the project exceeded expectations by offering a scalable framework that could easily be extended to more complex urban environments or integrated with live sensor data in future implementations.

8.3 Reflection and Skills Learned

This project has been a substantial learning experience, contributing to both technical and soft skill development. On the technical side, I acquired advanced skills in simulation modelling using SUMO, Python scripting for dynamic scenario generation, and data visualization using Jupyter Notebook. The project also deepened my understanding of intelligent transportation systems, adaptive signal control strategies, and environmental data integration within urban mobility contexts.

Beyond technical competencies, significant improvements were made in project management, academic writing, and critical analysis. Managing a multi-phase simulation project with evolving objectives required strong organizational skills and adaptability. Reviewing a wide range of academic literature developed my ability to synthesize complex information into coherent arguments, a skill that will be valuable in future research and professional endeavours.

If I were to undertake a similar project again, I would aim to incorporate real-time sensor data streams to enhance the project's responsiveness and explore machine learning-based adaptive control models. Additionally, I would allocate more time for iterative testing to fine-tune the balance between pedestrian and vehicular priorities under varying environmental scenarios. Overall, the project

reinforced the importance of modular design, early prototyping, and flexible planning when addressing real-world challenges.

8.4 Recommendations

While the project achieved its intended objectives, several opportunities for further development and optimization exist. Future work could focus on integrating live weather APIs or IoT-based rain sensors to enable real-time environmental responsiveness within the simulation or actual urban deployments. Scaling the simulation to model entire city networks, rather than a single intersection, would offer a broader understanding of system-wide impacts and challenges.

There is also potential to enhance the project through machine learning. By training reinforcement learning agents on diverse rainfall and traffic conditions, the system could evolve from rule-based adjustments to predictive, self-optimizing behaviour. Additional extensions could include developing priority signalling not only for pedestrians but also for emergency vehicles, cyclists, and public transport, further enhancing multi-modal traffic management.

Finally, improving the system's user-centric features such as auditory signals for visually impaired pedestrians or mobile notifications for crossing advisories could promote accessibility and inclusivity, aligning the project more closely with smart city initiatives.

In conclusion, this project lays a solid foundation for future innovations in environment-aware, pedestrian-friendly traffic control systems, contributing meaningfully to the advancement of safer, smarter urban mobility.

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10 Appendix

10.1 Guide to Run Simulations

To run the different simulations; after downloading the zip file on any computer, the baseline, rain scenario and rain-adaptive simulation configurations are located in config file. Make sure you already installed SUMO software on your computer before opening any of the simulations.

10.2 Ethics Training Certificate

This certificate below was required for this project.



Fig 10.1 Ethics Certificate

10.3 Rain Scenario Generation Script

```
import os
import csv
```

```

import xml.etree.ElementTree as ET
from datetime import datetime
import random

# =====
# PROJECT STRUCTURE
# =====
PROJECT_ROOT =
os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
DATA_DIR = os.path.join(PROJECT_ROOT, "data")
BASELINE_SCENARIO = os.path.join(PROJECT_ROOT, "scenarios",
"baseline")
RAIN_SCENARIO = os.path.join(PROJECT_ROOT, "scenarios", "rain")
HISTORICAL_RAIN_DATA = r"C:\Users\USER\OneDrive\Desktop\ICT\Major
project\project simulations\data\weather\historical_rain_data.csv"

# =====
# CONFIGURATION
# =====
RAIN_INTENSITY = {
    "none": {"ped_duration_range": (25, 35), "veh_speed": 13.89},
    "light": {"ped_duration_range": (40, 55), "veh_speed": 10.0},
    "moderate": {"ped_duration_range": (55, 70), "veh_speed": 8.0},
    "heavy": {"ped_duration_range": (80, 100), "veh_speed": 5.0}
}

def get_rain_intensity(rainfall):
    if rainfall < 0.1:
        return "none"
    elif 0.1 <= rainfall < 5.0:
        return "light"
    elif 5.0 <= rainfall < 15.0:
        return "moderate"
    else:
        return "heavy"

def generate_rain_scenario_routes():
    # Check file paths
    if not os.path.exists(HISTORICAL_RAIN_DATA):
        raise FileNotFoundError(f"Rain data file not found at:
{HISTORICAL_RAIN_DATA}")
    if not os.path.exists(BASELINE_SCENARIO):
        raise FileNotFoundError(f"Baseline scenario folder not found
at: {BASELINE_SCENARIO}")
    os.makedirs(RAIN_SCENARIO, exist_ok=True)

    # Read rainfall data
    max_rainfall = 0.0
    with open(HISTORICAL_RAIN_DATA, "r") as f:
        reader = csv.DictReader(f)
        for row in reader:

```

```

        try:
            rainfall = float(row["Rainfall (mm/h)"])
            if rainfall > max_rainfall:
                max_rainfall = rainfall
        except (ValueError, KeyError):
            continue

    # Get rain intensity and effects
    intensity = get_rain_intensity(max_rainfall)
    effects = RAIN_INTENSITY[intensity]
    ped_min, ped_max = effects["ped_duration_range"]
    veh_speed = effects["veh_speed"]

    print(f"[INFO] Max rainfall observed: {max_rainfall} mm/h")
    print(f"[INFO] Rain intensity: {intensity.upper()}")
    print(f"[INFO] Pedestrian walk duration: {ped_min}-{ped_max}
sec")
    print(f"[INFO] Vehicle max speed: {veh_speed} m/s")

    # ===== MODIFY VEHICLES =====
    veh_in = os.path.join(BASELINE_SCENARIO, "vehicles.rou.xml")
    veh_out = os.path.join(RAIN_SCENARIO, "vehicles.rou.xml")
    veh_tree = ET.parse(veh_in)
    for vType in veh_tree.findall("./vType"):
        vType.set("maxSpeed", str(veh_speed))
    veh_tree.write(veh_out)

    # ===== MODIFY PEDESTRIANS =====
    ped_in = os.path.join(BASELINE_SCENARIO, "pedestrians.rou.xml")
    ped_out = os.path.join(RAIN_SCENARIO, "pedestrians.rou.xml")
    ped_tree = ET.parse(ped_in)

    for person in ped_tree.findall("./person"):

        # Randomize walk duration within range
        walk_elem = person.find("walk")
        if walk_elem is not None:
            duration = random.randint(ped_min, ped_max)
            walk_elem.set("duration", str(duration))

    ped_tree.write(ped_out)

    print(f"[SUCCESS] Rain scenario files written to:
{RAIN_SCENARIO}")

if __name__ == "__main__":
    generate_rain_scenario_routes()

```

10.4 Rain-adaptive Scenario Generation Script

```

import os
import csv

```

```

import xml.etree.ElementTree as ET
from datetime import datetime
import random

# ===== CONFIGURABLE PROJECT STRUCTURE ===== #
PROJECT_ROOT =
os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
DATA_DIR = os.path.join(PROJECT_ROOT, "data")
BASELINE_SCENARIO = os.path.join(PROJECT_ROOT, "scenarios",
"baseline")
RAIN_SCENARIO = os.path.join(PROJECT_ROOT, "scenarios", "rain
adaptive")
HISTORICAL_RAIN_DATA = os.path.join(DATA_DIR, "weather",
"historical_rain_data.csv")

# ===== RAIN INTENSITY RULES ===== #
RAIN_INTENSITY = {
    "none": {"ped_duration": 30, "veh_speed": 13.89},
    "light": {"ped_duration": 45, "veh_speed": 11.5},
    "moderate": {"ped_duration": 60, "veh_speed": 9.0},
    "heavy": {"ped_duration": 90, "veh_speed": 7.0}
}

def get_rain_intensity(rainfall):
    """Classify rain intensity based on mm/h values"""
    if rainfall < 0.1:
        return "none"
    elif rainfall < 5.0:
        return "light"
    elif rainfall < 15.0:
        return "moderate"
    else:
        return "heavy"

def generate_rain_adaptive_scenario():
    # ==== Check project structure ====
    if not os.path.exists(HISTORICAL_RAIN_DATA):
        raise FileNotFoundError(f"Missing rainfall data file:
{HISTORICAL_RAIN_DATA}")
    if not os.path.exists(BASELINE_SCENARIO):
        raise FileNotFoundError(f"Missing baseline scenario:
{BASELINE_SCENARIO}")
    os.makedirs(RAIN_SCENARIO, exist_ok=True)

    # ==== Read worst-case rainfall value ====
    max_rainfall = 0.0
    with open(HISTORICAL_RAIN_DATA, "r") as f:
        reader = csv.DictReader(f)
        for row in reader:
            rainfall = float(row["Rainfall (mm/h)"])
            if rainfall > max_rainfall:

```

```

        max_rainfall = rainfall

    rain_type = get_rain_intensity(max_rainfall)
    ped_time = RAIN_INTENSITY[rain_type]["ped_duration"]
    veh_speed = RAIN_INTENSITY[rain_type]["veh_speed"]
    print(f"Detected Rainfall Intensity: {rain_type.upper()} → Ped
Duration: {ped_time}s, Vehicle Speed: {veh_speed} m/s")

    # ==== Modify vehicle route file ====
    vehicle_file = os.path.join(BASELINE_SCENARIO,
"vehicles.rou.xml")
    output_vehicle_file = os.path.join(RAIN_SCENARIO,
"vehicles.rou.xml")
    tree = ET.parse(vehicle_file)
    for vType in tree.findall("./vType"):
        vType.set("maxSpeed", str(veh_speed)) # adjust vehicle
speed for safety
    tree.write(output_vehicle_file)

    # ==== Modify pedestrian file ====
    pedestrian_file = os.path.join(BASELINE_SCENARIO,
"pedestrians.rou.xml")
    output_pedestrian_file = os.path.join(RAIN_SCENARIO,
"pedestrians.rou.xml")
    tree = ET.parse(pedestrian_file)
    root = tree.getroot()

    for person in root.findall("./person"):
        walk = person.find("walk")
        if walk is not None:
            walk.set("duration", str(ped_time)) # give longer
crossing time

    tree.write(output_pedestrian_file)
    print(f" Rain-adaptive scenario generated in: {RAIN_SCENARIO}")

if __name__ == "__main__":
    generate_rain_adaptive_scenario()

```